Design Evaluation 3 and Reflection

Duolingo XR — Immersive Chinese Learning

Chinese VR Learning Prototype — IP2b

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Objective and Validation Metrics

The prototype explores whether learners with limited prior Chinese knowledge can connect Chinese vocabulary with objects and actions in a virtual study room. IP2b develops the earlier card-and-selection experience through a moving-object task: participants catch a red book, place it inside a school bag and release it. The core sequence was introduction, six vocabulary cards, a ready screen, Task 1, Task 2 and completion feedback. Controller guidance was added before the 23 October sessions. The words are 书 (book), 电脑 (computer), 书包 (school bag), 找到 (find), 拿起 (pick up) and 放进 (put into). Cards provide Chinese characters, pinyin, English meanings and recorded pronunciation. Room objects no longer open additional vocabulary cards.

The evaluation examined whether participants could discover the task sequence, distinguish selection from grabbing, and understand placement and release. The test plan proposed recording task time, errors and assistance separately from vocabulary scores. Its provisional usability target was four of five participants independently completing both tasks, rather than a course grading threshold. Before and after the VR experience, I pronounced the six Chinese words and asked participants to explain their meanings. Each correct meaning received one point. This assessed immediate understanding of spoken words outside VR, rather than written-character recognition or comprehension during gameplay. Higher scores would indicate immediate change, not establish that the interaction caused learning.

Research provides a basis for interpreting the evaluation. Legault et al., (2019) found higher Mandarin recognition accuracy in immersive VR than in word–word paired-association learning. Macedonia et al., (2023) found an additional vocabulary-retention benefit from virtual grasping among lower language-aptitude learners in a VR CAVE. These studies support exploring contextual interaction, although neither establishes the effectiveness of this Quest prototype. CAMIL identifies presence, agency and cognitive load as relevant learning processes(Makransky & Petersen, 2021) . Kao et al., (2021) found that controller-tutorial effectiveness varied across game types, supporting attention to control instructions. Nielsen & Landauer's (1993) problem-discovery model provides a basis for small, iterative usability evaluations; it does not establish that five participants can demonstrate learning effectiveness.

Results

Five participants tested the prototype across two sessions. Two reported no Chinese experience; three reported prior exposure, with P5's level unspecified. Baseline scores ranged from 0/6 to 3/6, so this was not a sample of five complete beginners. All five completed Task 1 and automatically entered Task 2. P1 received one H1 prompt; P2–P5 had no timed-task assistance recorded, although P3 needed an earlier prompt to find Start Task. Task 1 durations were 23–58 seconds and Task 2 durations were 41–116 seconds (Appendix A, Tables A1–A2). Final Task 2 outcomes remain unconfirmed. P1 and P2 received grab-control guidance, P3 received three prompts, and P1, P2 and P4 held the book inside the bag without releasing it. All five improved in the live oral vocabulary checks, from a mean of 1.4/6 to 5.2/6. All identified 电脑, 书包 and 找到 afterwards; three identified 拿起 (Tables A3–A4). These changes describe immediate spoken word-meaning performance in a sample with mixed prior knowledge.

Mean ratings were 4.0/5 for knowing the next step, 4.2/5 for computer selection, 3.6/5 for grabbing and placement, 5.0/5 for feedback, and 4.6/5 for perceived vocabulary support (Table A5). Observations nevertheless recorded difficulty finding Start Task, recognising the target bag and understanding release. P4 described grabbing as enjoyable; P3 requested clearer visual cues, and P5 suggested removing the competing black bag (Table A6).

Analysis

Insight 1: The next action was not sufficiently clear after entering the room.

P1 was unsure what to do, while P3 explored the room before finding Start Task following a prompt. Completing the vocabulary cards did not necessarily prepare participants to begin the tasks independently. Although everyone rated understanding the next step at 4/5, the observations revealed a

specific navigation problem. The start button should be placed within the initial viewing area and supported by a clear label or colour cue.

Insight 2: Interaction difficulties should not automatically be interpreted as language difficulties.

P1 and P2 needed grab-control guidance, so controller knowledge contributed to their difficulties. Although the plan included a pen-based demonstration of selection, grabbing and release, I did not conduct it before testing. This departure made language and control difficulties harder to separate. Controller instructions were added before the later sessions, but participant differences prevent attributing task-time differences to this change alone. Future sessions should implement the planned familiarisation and check each action before the language tasks. Kao et al., (2021) also identify tutorial format as relevant to control learning, with effects varying across VR game types.

Insight 3: Placement did not clearly communicate the remaining release requirement.

P1, P2 and P4 held the book inside the bag without releasing it. Participants may have interpreted the book's visible position as completion, while the system required release. P3's repeated placements and P5's confusion between bags also indicate uncertainty about the target area and object. Feedback should distinguish these stages: identifying the target bag, entering the accepted area and releasing the book. An interior highlight and a brief release prompt could communicate progress before the final completion message.

Insight 4: Immediate vocabulary performance improved, but the contribution of grabbing remains uncertain.

All five participants improved in the oral checks, with the mean increasing from 1.4/6 to 5.2/6. However, only three correctly identified 拿起 after the experience. Performing a grabbing action therefore did not ensure understanding of its Chinese label. Participants also encountered English, pinyin, cards and pronunciation audio, whose contributions cannot be separated here. Macedonia et al., (2023) and Legault et al., (2019) provide a rationale for further investigation, but this study did not compare learning methods. A future task could use the same action word with a different object to examine whether understanding transfers beyond the original book task.

Evaluation against the five lecture design principles

Principle	Evidence and judgement	Proposed response
Text Readability	Feedback ratings were positive, but they combine sight, sound and understanding. No separate readability score was recorded.	Check Chinese glyphs, contrast, size and occlusion on the headset; test instruction reading separately.
Ergonomics	Participants performed a reach-and-place task, but systematic reach and fatigue data are absent.	Keep the book and bag within comfortable reach and observe overextension or sustained arm elevation.
Brightness Changes	The records do not document brightness transitions.	Inspect interface and feedback transitions for abrupt changes and ask about visual discomfort.
Avoiding simulator sickness	The records do not provide a dedicated sickness measure.	Assess discomfort before and after use and monitor movement and viewpoint changes.
Button Placement	P3 missed Start Task and needed a prompt.	Reposition and visually distinguish the start control, then retest discovery without assistance.

Evaluation of Aims

The evaluation partially supported the aims. All five completed Task 1; four had no facilitator assistance recorded during the timed task. However, P3 needed earlier navigation help, and Task 2 outcomes were

not confirmed. The target of four participants independently completing both tasks therefore cannot be assessed reliably. Placement, release and target identification remained barriers.

All five improved in the oral vocabulary checks, but this does not establish understanding during VR gameplay, long-term retention or a specific benefit of manipulation. Prior knowledge varied, and live pronunciation departed from the planned standardised recordings. The 4.6/5 perceived-learning rating indicates a positive subjective response rather than established learning effectiveness.

Concept Iteration

The next iteration would retain the sequence of vocabulary learning followed by object-based practice, while prioritising clearer interaction. Start Task would be placed within the initial viewing area with an explicit cue. Controller diagrams or a short practice activity would explain grabbing and release. The competing black bag would be removed or clearly distinguished, and a release prompt would appear when the red book enters the accepted area. These changes address the observed problems with task discovery, controls and completion conditions. The evaluation method would also be revised. A future VR task could ask participants to follow spoken Chinese instructions using different objects, examining whether understanding transfers to a new context. A delayed vocabulary check would assess retention. Participants would use the same build, with action errors, assistance levels and final outcomes recorded separately. This would help distinguish learning the game's controls from understanding the Chinese vocabulary.

Final Reflection

Developing IP2b shifted the project from vocabulary cards and object selection towards physical language practice. The moving-book task made interaction more central, but introduced difficulties beyond Chinese comprehension. P1 and P2 needed grab-control guidance, while P1, P2 and P4 held the book inside the bag without releasing it. I had assumed that placement and release would feel like one familiar action. Testing showed that the interface needed to communicate this distinction.

Combining observations and ratings was useful: all participants rated feedback at 5/5, yet some were unsure about release. My procedure had two departures from the plan. I pronounced vocabulary live instead of using identical recordings, potentially varying pronunciation, pacing and loudness. I also omitted the planned pen-based control demonstration. This weakened standardisation and made language and control difficulties harder to separate. Incomplete outcome and assistance records further limited interpretation. Changing controller instructions supported iteration, but did not provide a controlled comparison. The records do not establish whether a separate pilot was conducted.

The concept was partially supported. Participants viewed manipulation positively and all improved in immediate oral checks. However, prior knowledge varied, and checks outside VR did not establish in-game comprehension, retention or the specific benefit of grabbing. Completing an activity and demonstrating learning require different evidence.

Next, I would improve task-start guidance, bag identification and release feedback. Future tests would implement familiarisation, use standardised recordings and a consistent build, and record errors, assistance and final outcomes separately. New objects could test transfer, followed by a delayed vocabulary check. Hand tracking could reduce button-learning demands, but tracking, release recognition and the loss of controller vibration require evaluation. I would compare it with controllers before adoption. XR techniques should address observed needs, with learning assessed separately from technical appeal.

GitHub Link

https://github.com/yenyong1211/YININGZHU_DECO7230_2026/tree/main/Prototype2a

https://github.com/yenyong1211/YININGZHU_DECO7230_2026/tree/main/Evaluation_3

References

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Appendix

Appendix A — Detailed Test Results

Table A1. Participant Background and Recorded Stage Durations

Participant	Test Date	Chinese Experience	VR Experience	Learning Cards (s)	Task 1 (s)	Task 2 (s)
P1	21 October	None	None	137	58	113
P2	21 October	None	A little	81	23	105
P3	23 October	A little	Frequent use	39	47	116
P4	23 October	A few previously learned words	A little	33	29	41
P5	23 October	Some experience	Some experience	47	25	112

Note. Dates refer to 2026. Controller instructions were added for the 23 October sessions. Durations are recorded stage times, not consistently verified successful-completion times. Complete session durations were not recorded.

Table A2. Task Outcomes, Interaction Errors and Facilitator Assistance

Participant	Task 1 Outcome and Errors	Task 1 Assistance	Task 2 Outcome and Errors	Task 2 Assistance
P1	Initially selected the telephone; two errors recorded. Final outcome unclear.	One H1 prompt.	Two outside releases and one held-book wait described; numeric error total entered as two. Final outcome unclear.	Grab and release guidance, consistent with H2. Prompt count unspecified.
P2	Initially selected the table; one error recorded. Final outcome unclear.	H0: no assistance recorded.	Difficulty grabbing; one held-book wait recorded. Final outcome unclear.	Grab-control explanation, consistent with H2. Prompt count unspecified.
P3	Recorded as successful; zero errors.	H0 during the task. A separate navigation prompt helped locate Start Task beforehand.	Marked as failed; two inaccurate placements. Final task endpoint unclear.	Three prompts; levels unspecified.
P4	Recorded as successful; error count blank.	H0: no assistance recorded.	Marked as failed; one held-book wait. Whether later release produced completion is unclear.	Not clearly recorded.
P5	Recorded as successful; zero errors.	H0: no assistance recorded.	Marked as failed; initially chose the wrong bag. Later correct placement described, but system completion unconfirmed.	Retry prompt, consistent with H1. Exact count unspecified.

Note. H0 = no facilitator assistance; H1 = neutral prompt or facilitator-provided replay; H2 = operational guidance; H3 = direct answer or demonstration. Unless explicitly entered in the original record, levels are retrospective classifications based on the described help. Unspecified prompts remain unclassified. Incorrect actions are not automatically classified as task failure.

Table A3. Individual Oral Vocabulary Scores Before and After the Experience

Participant	Before / 6	After / 6	Change
P1	0	4	+4
P2	0	5	+5
P3	2	5	+3
P4	2	6	+4
P5	3	6	+3
Mean	**1.4**	**5.2**	**+3.8**

Note. The facilitator pronounced the words live before and after the VR experience and checked their meanings. Each correct meaning received one point. Scores assess immediate understanding of spoken words, not written-character recognition or long-term retention.

Table A4. Correct Responses by Vocabulary Item

Chinese Word	English Meaning	Correct Before (n/5)	Correct After (n/5)
书	Book	3/5	4/5
电脑	Computer	1/5	5/5
书包	School bag	3/5	5/5
找到	Find	0/5	5/5

拿起	Pick up	0/5	3/5
放进	Put into	0/5	4/5

Note. Values indicate the number of participants who correctly identified each meaning. Vocabulary understanding was checked outside the VR tasks.

Table A5. Participant Ratings

Statement	P1	P2	P3	P4	P5	Mean
I understood what to do next.	4	4	4	4	4	4.0
I could select the computer easily.	3	4	5	4	5	4.2
I could grab and place the red book in the bag.	4	3	4	3	4	3.6
I could see, hear and understand the feedback.	5	5	5	5	5	5.0
Manipulating objects helped me understand the Chinese words.	4	5	5	4	5	4.6

Note. Ratings range from 1, strongly disagree, to 5, strongly agree. Statements were rated individually and do not form a validated composite scale. Perceived learning support is not an objective measure of learning effectiveness.

Table A6. Observations and Participant Suggestions

Participant	Observation or Feedback	Participant Suggestion
P1	Was unsure what to do after entering the room.	No specific suggestion recorded.
P2	Did not know which controller control to use for grabbing.	Add controller instructions.
P3	Explored the room before locating Start Task following a prompt.	Add colour cues and clearer animation when the book is placed inside the bag.
P4	Described grabbing as enjoyable but did not know the book had to be released inside the bag.	No separate suggestion recorded.
P5	Mistook the black bag for the target and overlooked the bag on the desk.	Remove the competing black bag.

Note. Entries are English paraphrases of the original Chinese notes, not verbatim quotations. The records state that controller instructions were added for the later sessions.